

ANDRÉS LAGE FREIRE

Responsible AI Architect | AI Governance & AI Assurance

Spain · Open to EMEA / Remote

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EXECUTIVE SUMMARY

Responsible AI Architect and Strategic AI Governance Lead specialized in helping organizations transform AI initiatives from proof-of-concept into governed, auditable and scalable systems. Operating at the intersection of business, engineering, cybersecurity, legal and compliance functions, I establish AI Governance operating models, risk management frameworks and Responsible AI controls aligned with the EU AI Act, ISO/IEC 42001 and NIST AI RMF — translating regulatory obligations into executable controls, SDLC gates and runtime assurance.

SPECIALIZED IN

AI Governance	Responsible AI	Agentic AI Governance
EU AI Act Readiness	ISO/IEC 42001	AI Risk Management
AI Assurance	Enterprise AI Architecture	AI Lifecycle Governance

SELECTED AI GOVERNANCE PROJECTS

Enterprise AI Vendor Governance Console

github.com/tshapedconsultant/enterprise-ai-risk

STACK · V1.4.0

- Deterministic scoring · GDPR + EU AI Act + ISO/IEC 42001 + NIST AI RMF
- Jira human gates · hash-chained audit (Jira/Rekor/S3)
- FastAPI · Pytest · Docker · GitHub Actions CI

Outcome. Executable vendor risk controls: deterministic rules engine with HMAC-validated webhooks, external audit anchoring, and CI-deployed evidence packs.

Responsible AI Compliance Hub

github.com/tshapedconsultant/porto-seguro-compliance-hub

DATASET · 595K RECORDS

- EBM glass-box classifier · Polars ingestion
- KS drift gates · fairness audits (Art. 9, 14, 15)
- JSON evidence packs · Solvency II conformity · FastAPI · Streamlit

Outcome. Five-layer reference architecture mapping EU AI Act Articles 9–15 to executable controls; demonstration model achieved ROC-AUC 0.64.

Enterprise Data Analyst Agent

github.com/tshapedconsultant/Enterprise-Data-Analyst-Agent

STACK · 172 TESTS

- LangGraph supervisor-worker · deterministic routing — LLM never chooses next node by default
- 3-layer defense: prompt-injection guard · AST Python sandbox · SQL AST validation
- 172 tests · GDPR / EU AI Act gap analysis · FastAPI · LangChain · OpenAI GPT-4o

Outcome. Governance as invariant: the graph owns flow, the LLM produces and explains, deterministic rules route and retry.

Additional projects. AI Red Team Orchestrator — Azure-hosted GARAK framework for adversarial testing. tshapedconsultant.com — AI governance engineering website built with Cursor and LLM-assisted coding.

PROFESSIONAL EXPERIENCE

Strategic AI Consultant & Responsible AI Architect (Independent)

Jun 2021 – Present

International · Jun 2021 – Present

AI GOVERNANCE, RISK & STRATEGY

- Created DGOM™, a Board-to-Code AI governance and risk operating model translating regulatory obligations, risk appetite and executive decisions into executable controls, SDLC gates, runtime assurance and auditable evidence.
- Established Human-in-the-Loop (HITL), escalation and dual-approval patterns for high-impact AI decisions; benchmarked risk operating models from Microsoft, Deloitte and BBVA.

APPLIED AI ARCHITECTURE & TECHNICAL LEADERSHIP

- Architected governed multi-agent AI prototypes using LangGraph, FastAPI and OpenAI API with control-by-design principles.
- Designed controls-first ML pipelines integrating explainability, bias monitoring, drift detection and tamper-evident audit logging using PyTorch, TensorFlow and Scikit-learn.
- Conducted a statistically rigorous Pandas vs Polars benchmark (8 warmups, 20 runs, CV analysis) demonstrating 5.5× faster CSV processing.
- Stakeholder Leadership. Acted as trusted advisor between engineering, compliance, cybersecurity, legal and executive stakeholders.

Cloud Implementation Specialist — Avaya Cloud Office

Oct 2020 – Jun 2021

Concentrix · Sofia, Bulgaria

- Led cloud deployment projects for multinational enterprise clients under strict security and compliance requirements.
- Acted as primary interface between technical delivery teams and executive stakeholders; streamlined rollout processes for multinational enterprise clients.

Innovation Project Lead (Award Winner)

2018

Repsol Foundation · 2018

- Led the team awarded the Repsol Foundation Entrepreneurship Prize.
- Applied Lean experimentation and business validation methodologies with corporate partners.

CORE COMPETENCIES

AI GOVERNANCE & RISK	AGENTIC AI & LLM ENGINEERING	ML/DL ENGINEERING
• EU AI Act • ISO/IEC 42001 • NIST AI RMF • AI Risk Assessment • Human-in-the-Loop Governance • AI Assurance • AI Red Teaming	• Multi-Agent Systems • LangGraph • LangChain • RAG Architectures • OpenAI API • Azure OpenAI • GARAK	• PyTorch • TensorFlow • Scikit-learn • Model Deployment • FastAPI • Monitoring & Validation
CLOUD, DATA & MLOPS	SECURITY, TRUST & COMPLIANCE	LEADERSHIP
• Python • SQL • Docker • Kubernetes • Polars • Streamlit • Azure AI Services	• GDPR • AI Compliance • Explainability (SHAP, LIME) • Prompt Injection Defense • Vendor Risk • Tamper-evident Audit	• Executive Communication • Stakeholder Alignment • Strategic Advisory • AI Lifecycle Transformation

EDUCATION & CREDENTIALS

Harvard SEAS

2026

Agentic AI Foundations: Business Applications and Risks

Harvard Online

2026

Innovation Strategy: Tools and Frameworks for Business

Stanford Online

2026

Cybersecurity and Executive Strategy (XACS302)

Stanford Online

2026

How Software Ate Finance · Stanford GSB

Oxford Saïd · UNESCO

2026

AI, Justice, and the Rule of Law

IAPP AIGP

2026

Examination scheduled 30 November 2026 · Certification in progress

Oxford — Saïd Business School

2025

AI Governance Programme

Wharton

2025

AI Strategy & Governance

Wharton Online

2025

Fintech: Foundations & Applications of Financial Technology (Specialization)

Independent AI Governance & AI Systems Research Programme (2025–Present)
Independent research and applied study spanning AI governance, agentic AI, ML/MLOps, cybersecurity, privacy and Responsible AI, validated through implementation and comparative analysis.

Additional programmes: Stanford Advanced Learning Algorithms; Michigan Responsible Generative AI; Yale Narrative Economics.

“Rapid learning is the key to success.”

Eric Schmidt, former CEO of Google · cited by Salim Ismail and Peter Diamandis, Moonshots (ExO 3.0)

Chemical Engineering (Plan Superior)

2002–2008

IQS School of Engineering – Universitat Ramon Llull · 2002–2008 · 247.5 ECTS